



Thi Ngoc Trang Tran

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RESEARCH DOMAINS

Recommender Systems, Group Recommender Systems, Human Decision Making, Intelligent User Interfaces, Explanations, Configuration Systems, Software Engineering

EDUCATION

2015 - 2019	Ph.D. student at the Graz University of Technology, Graz, Austria
2006 - 2008	Master student at the Faculty of Computer Science, College of Sciences, Hue University, Vietnam
2000 - 2004	Bachelor student at the Faculty of Computer Science, College of Sciences, Hue University, Vietnam

JOBS

since 2019	Senior Postdoctoral Researcher, University Assistant with Doctorate at the Institute of Software Engineering and AI, Graz University of Technology, Graz, Austria
2015 - 2019	Awarded a full Vietnamese–Austrian Scholarship to pursue a Ph.D. at Graz University of Technology, Austria.
2009 - 2015	Lecturer at the Department of Information Technology and Communication, the School of Hospitality and Tourism, Hue University, Vietnam
2004 - 2008	Project assistant at the Center of Information Technology, Hue University, Vietnam

GRANTS & AWARDS

2023 & 2024	Nominated for the best teaching in the course “Introduction to Structured Programming”
2019	ACM UMAP 2019 - SIGCHI Student Travel Grant
2018	Best Paper Award - Configuration Workshop 2018 (CONFWS’ 2018)
2017	Best Paper Award - SEKE’ 2017
2016	Runner-up Best Paper Award - Configuration Workshop 2016 (CONFWS’ 2016)
2015 - 2019	Vietnamese - Austrian Doctoral Grant (sponsored by the Vietnamese Government and the OeAD organization - Austria)

- 2004 Awarded recognition among the top Bachelor's graduates in Computer Science with excellent and outstanding academic results.
- 2003 Odon-Vallet Grant for excellent Computer Science students

PUBLICATIONS

Journal papers

- J13 A. Felfernig, M. Wundara, **T. N. T. Tran**, V. M. Le, S. Lubos, and S. Polat-Erdeniz (2024). Sports recommender systems: overview and research directions, *Journal of Intelligent Information Systems*. <https://doi.org/10.1007/s10844-024-00857-w>
- J12 M. Uta, A. Felfernig, V.M. Le, **T. N. T. Tran**, D. Garber, S. Lubos, and T. Burgstaller (2024). Knowledge-based Recommender Systems: Overview and Research Directions. *Frontiers in Big Data*, section Recommender Systems. <https://doi.org/10.3389/fdata.2024.1304439>
- J11 A. Felfernig, M. Wundara, **T. N. T. Tran**, S. Polat-Erdeniz, S. Lubos, M. El-Mansi, D. Garber, and V. M. Le (2023). Recommender Systems for Sustainability: Overview and Research Issues. *Frontiers in Big Data*, section Recommender Systems. <https://doi.org/10.3389/fdata.2023.1284511>
- J10 **T. N. T. Tran**, A. Felfernig, and V.M. Le (2023). An Overview of Consensus Models for Group Decision-Making and Group Recommender Systems. *Journal of User Modeling and User-Adapted Interaction*. <https://doi.org/10.1007/s11257-023-09380-z>
- J9 S. P. Erdeniz, S.P.K. Veeranki, S. Jauk, **T. N. T. Tran**, A. Felfernig, M. Schrempf, D. Kramer, and W. Leodolter (2022). Explaining Predictions of Decision Support Systems in Healthcare. *Current Directions in Biomedical Engineering*, vol. 8, no. 2, pp. 117-120. <https://doi.org/10.1515/cdbme-2022-1031>
- J8 E. Lex, D. Kowald, P. Seitlinger, **T. N. T. Tran**, A. Felfernig, and M. Schedl (2021). Psychology informed Recommender Systems. *Foundations and Trends® in Information Retrieval*, vol. 15, no. 2, pp 134-242. <http://dx.doi.org/10.1561/15000000090>
- J7 **T. N. T. Tran**, A. Felfernig, and N. Tintarev (2021). Humanized Recommender Systems: State-of-the-art and Research Issues. *ACM Transactions on Interactive Intelligent Systems*, vol. 11, no. 2, article 9, pp. 1-41. <https://doi.org/10.1145/3446906>
- J6 V. M. Le, A. Felfernig, **T. N. T. Tran**, M. Atas, M. Uta, D. Benavides, and J. Galindo (2021). DirectDebug: A Software Package for the Automated Testing and Debugging of Feature Models. *Software Impacts*, 100085, vol. 9. <https://doi.org/10.1016/j.simp.2021.100085>
- J5 M. Atas, A. Felfernig, S. Polat-Erdeniz, A. Popescu, **T. N. T. Tran**, and M. Uta (2021). Towards psychology-aware preference construction in recommender systems: Overview and research issues. *Journal of Intelligent Information Systems*. <https://doi.org/10.1007/s10844-021-00674-5>
- J4 **T. N. T. Tran**, A. Felfernig, C. Trattner, and A. Holzinger (2020). Recommender systems in the healthcare domain: state-of-the-art and research issues. *Journal of Intelligent Information Systems*. <https://doi.org/10.1007/s10844-020-00633-6>
- J3 **T. N. T. Tran**, M. Atas, V.M. Le, R. Samer, and M. Stettinger (2020). Social Choice-based Explanations: An Approach to Enhancing Fairness and Consensus Aspects. *Journal of Universal Computer Science*, vol. 26, no. 3, pp. 402-431.
- J2 S. Polat-Erdeniz, A. Menychtas, I. Maglogiannis, A. Felfernig, and **T. N. T. Tran** (2019). Recommender systems for IoT enabled quantified-self applications. *Evolving Systems*, vol. 11, pp. 291-304. <https://doi.org/10.1007/s12530-019-09302-8>
- J1 **T. N. T. Tran**, M. Atas, A. Felfernig, and M. Stettinger (2018). An Overview of Recommender Systems in the Healthy Food Domain. *Journal of Intelligent Information Systems (JIIS)*, vol. 50, pp. 501-526. <https://doi.org/10.1007/s10844-017-0469-0>

Conference papers

- C21 **T. N. T. Tran**, S. P. Erdeniz, A. Felfernig, S. Lubos, M. E. Mansi, and V. M. Le (2024). Less is More: Towards Sustainability-Aware Persuasive Explanations in Recommender Systems, In Proceedings of the 18th ACM Conference on Recommender Systems - RecSys 2024. Association for Computing Machinery (ACM). <https://doi.org/10.1145/3640457.3691708>
- C20 S. Lubos, **T. N. T. Tran**, A. Felfernig, S. P. Erdeniz, and V. M. Le (2024). LLM-generated Explanations for Recommender Systems, UMAP 2024 - Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization. Association for Computing Machinery (ACM). <https://doi.org/10.1145/3631700.36651>
- C19 S. Lubos, A. Felfernig, **T. N. T. Tran**, D. Garber, E. Merfat, S. P. Erdeniz, and V. M. Le (2024). Leveraging LLMs for the Quality Assurance of Software Requirements, in proceedings of 32nd International Requirements Engineering Conference (RE), Reykjavik, Iceland, 2024, pp. 389-397, doi: 10.1109/RE59067.2024.00046.
- C18 V. M. Le, A. Felfernig, **T. N. T. Tran**, and M. Uta (2024). InformedQX: Informed Conflict Detection for Over-Constrained Problems. In 38th AAAI Conference on Artificial Intelligence. AAAI'24, Vancouver, Canada. <https://doi.org/10.1609/aaai.v38i9.28932>
- C17 T. Burgstaller, V. M. Le, **T. N. T. Tran**, and A. Felfernig (2023). FMTesting: A FeatureIDE Plug-in for Automated Feature Model Analysis and Diagnosis. In 12th Conference on Prestigious Applications of Intelligent Systems (PAIS 2023), Kraków, Poland. DOI: 10.3233/FAIA230640
- C16 S. Lubos, A. Felfernig, V. M. Le, **T. N. T. Tran**, D. Benavides, J.A. Zamudio, and D. Garber (2023). Analysis Operations On The Run: Feature Model Analysis in Constraint-based Recommender Systems. In 27th ACM International Systems and Software Product Line Conference (SPLC 2023), Tokyo, Japan. <https://doi.org/10.1145/3579027.3608982>
- C15 **T. N. T. Tran**, A. Felfernig, V. M. Le, T. M. N. Chau, and T. G. Mai (2023). User Needs for Explanations of Recommendations: In-depth Analyses of the Role of Item Domain and Personal Characteristics. In 31st ACM Conference on User Modelling, Adaptation and Personalization. ACM UMAP 2023, Limassol, Cyprus. <https://doi.org/10.1145/3565472.3592950>
- C14 V. M. Le, C.V. Silva, A. Felfernig, **T. N. T. Tran**, J. Galindo, and D. Benavides (2023). FastDiagP: An Algorithm for Parallelized Direct Diagnosis. In 37th AAAI Conference on Artificial Intelligence. AAAI'23, Washington, DC, USA. <https://doi.org/10.1609/aaai.v37i5.25792>
- C13 V.M. Le, A. Felfernig, M. Uta, **T. N. T. Tran**, and C. Vidal-Silva (2022). WipeOutR: Automated Redundancy Detection for Feature Models. In 26th ACM International Systems and Software Product Line Conference – Volume A (SPLC'22), Graz, Austria. <https://doi.org/10.1145/3546932.3546992>
- C12 **T. N. T. Tran**, V. M. Le, M. Atas, A. Felfernig, M. Stettinger, and A. Popescu (2021). Do Users Appreciate Explanations of Recommendations? An Analysis in the Movie Domain. In the RecSys 2021 Late-Breaking Results, pp. 645-650, Virtual. <https://doi.org/10.1145/3460231.3478859>
- C11 M. Uta, A. Felfernig, V. M. Le, A. Popescu, **T. N. T. Tran**, and D. Helic (2021). Evaluating Recommender Systems in Feature Model Configuration. In the 25th ACM International Systems and Software Product Line Conference (SPLC 2021), pp.58-63, Virtual. <https://doi.org/10.1145/3461001.3471144>
- C10 V. M. Le, A. Felfernig, M. Uta, D. Benavides, J. Galindo, and **T. N. T. Tran** (2021). DirectDebug: Automated Testing and Debugging of Feature Models. In 2021 IEEE/ACM 43rd International Conference on Software Engineering: New Ideas and Emerging Results (ICSE-NIER), pp. 81-85, IEEE/ACM, Virtual. <https://doi.org/10.1109/ICSE-NIER52604.2021.00025>

- C9 V. M. Le, **T. N. T. Tran**, and A. Felfernig (2021). A Conversion of Feature Models into an Executable Representation in Microsoft Excel. In: Stettinger M., Leitner G., Felfernig A., Ras Z.W. (eds) *Intelligent Systems in Industrial Applications* (pp. 153-168). ISMIS 2020, Graz, Austria. *Studies in Computational Intelligence*, vol 949. https://doi.org/10.1007/978-3-030-67148-8_12
- C8 M. Stettinger, **T. N. T. Tran**, I. Pribik, G. Leitner, A. Felfernig, R. Samer, M. Atas, and M. Wundara (2020). KNOWLEDGECHECKR: Intelligent Techniques for Counteracting Forgetting. In: G. DeGiacomo, A. Catala, B. Dilkina, M. Milano, S. Barro, A. Bugarin, J. Lang (eds.) *24th European Conference on Artificial Intelligence (ECAI 2020)*, Santiago de Compostela, Spain, vol. 325, pp. 3034-3039, *Frontiers in Artificial Intelligence and Applications*. IOS Press. <https://doi.org/10.3233/FAIA200479>
- C7 M. Atas, **T. N. T. Tran**, A. Felfernig, S. Polat Erdeniz, R. Samer, and M. Stettinger (2019). Towards Similarity-Aware Constraint-Based Recommendation. In: Wotawa F., Friedrich G., Pill I., Koitz-Hristov R., Ali M. (eds) *Advances and Trends in Artificial Intelligence. From Theory to Practice. International Conference on Industrial, Engineering and Other Applications of Applied Intelligent Systems (IEA/AIE 2019)*, Graz, Austria. *Lecture Notes in Computer Science*, vol 11606, pp. 287-299. Springer. https://doi.org/10.1007/978-3-030-22999-3_26
- C6 M. Atas, R. Samer, A. Felfernig, **T. N. T. Tran**, S. P. Erdeniz, and M. Stettinger. 2019. Socially-Aware Diagnosis for Constraint-Based Recommendation. In *Proceedings of the 27th ACM Conference on User Modeling, Adaptation and Personalization (UMAP '19)*. Association for Computing Machinery, New York, NY, USA, 121–129. <https://doi.org/10.1145/3320435.3320436>
- C5 **T. N. T. Tran**, M. Atas, A. Felfernig, V. M. Le, R. Samer, and M. Stettinger (2019). Towards Social Choice-based Explanations in Group Recommender Systems. In *Proceedings of the 27th ACM Conference on User Modeling, Adaptation and Personalization (UMAP '19)*, Larnaca, Cyprus, pp. 13-21. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3320435.3320437>
- C4 **T. N. T. Tran**, A. Felfernig, V. M. Le, M. Atas, M. Stettinger, and R. Samer (2019). User Interfaces for Counteracting Decision Manipulation in Group Recommender Systems. In *Adjunct Publication of the 27th Conference on User Modeling, Adaptation and Personalization (UMAP'19 Adjunct)*, pp. 93-98, Larnaca, Cyprus. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3314183.3324977>
- C3 **T. N. T. Tran**, M. Atas, A. Felfernig, R. Samer, and M. Stettinger (2018). Investigating Serial Position Effects in Sequential Group Decision Making. In *Proceedings of the 26th Conference on User Modeling, Adaptation and Personalization (UMAP '18)*, Singapore, pp. 239-243. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3209219.3209255>
- C2 M. Atas, S. Reiterer, A. Felfernig, **T. N. T. Tran**, and M. Stettinger (2018). Polarization Effects in Group Decisions. In *Adjunct Publication of the 26th Conference on User Modeling, Adaptation and Personalization (UMAP '18)*, Singapore, pp. 305-310. Association for Computing Machinery, New York, NY, USA, 305–310. <https://doi.org/10.1145/3213586.3225242>
- C1 M. Atas, **T. N. T. Tran**, A. Felfernig, and R. Samer (2018). Socially-aware Recommendation for Over-Constrained Problems. In: Mouhoub M., Sadaoui S., Ait Mohamed O., Ali M. (eds) *Recent Trends and Future Technology in Applied Intelligence. 31st International Conference on Industrial Engineering and Other Applications of Applied Intelligent Systems (IEA/AIE 2018)*, Montreal, Canada. *Lecture Notes in Computer Science*, vol 10868, pp. 267-278. Springer. https://doi.org/10.1007/978-3-319-92058-0_25

Workshop papers

- W17 S. Lubos, **T. N. T. Tran**, V. M. Le, D. Garber, M. Henrich, RR. Willfort, and J. Fuchs (2025). Towards LLM-Enhanced Group Recommender Systems, In *Proceedings of Joint Workshop on Interfaces and Human Decision Making for Recommender Systems*, September 22, 2025, Prague - IntRS'25, <https://ceur-ws.org/Vol-4027/paper5.pdf>

- W16 A. Felfernig, D. Garber, V. M. Le, S. Lubos, and **T. N. T. Tran** (2025). Sustainability Evaluation Metrics for Recommender Systems. in L. Boratto, F. M. Mallocci, A. De Filippo, E. Lex, N. Mauro, & F. Ricci (Hrsg.), *Recommender Systems for Sustainability and Social Good - Second International Workshop, RecSoGood 2025, Proceedings* (S. 14–26). (Communications in Computer and Information Science; Band 2802 CCIS). https://doi.org/10.1007/978-3-032-13342-7_2
- W15 A. Felfernig, D. Garber, V. M. Le, S. Lubos, and **T. N. T. Tran** (2025). Towards LLM-Enhanced Product Line Scoping. in *Proceedings of the 27th International Workshop on Configuration (ConfWS 2025) co-located with the 28th European Conference on Artificial Intelligence (ECAI 2025)* (Band 4149, S. 128-134). CEUR Workshop Proceedings. <https://ceur-ws.org/Vol-4149/paper11.pdf>
- W14 S. Lubos, A. Felfernig, D. Garber, V. M. Le, and **T. N. T. Tran** (2025). Towards LLM-Based Usability Analysis for Recommender User Interfaces. CEUR Workshop Proceedings, 4027. <https://ceur-ws.org/Vol-4027/paper7.pdf>
- W13 S.P. Erdeniz, **T. N. T. Tran**, A. Felfernig, D. Kramer, M. Schrempf, and P. Rainer (2023). Employing Nudge Theory and Persuasive Principles with Explainable AI in Clinical Decision Support. Workshop on Machine Learning and Artificial Intelligence in Bioinformatics and Medical Informatics (MABM 2023), in conjunction with the 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2023), Istanbul, Turkey. doi: 10.1109/BIBM58861.2023.10385315.
- W12 S. Lubos, **T. N. T. Tran**, S.P. Erdeniz, M.E. Mansi, A. Felfernig, M. Wundara, and G. Leitner (2023). Concentrating on Impact: Consequence-based Explanations in Recommender Systems. In 10th Joint Workshop on Interfaces and Human Decision Making for Recommender Systems (IntRS'23), co-located with 17th ACM Conference on Recommender Systems (RecSys 2023), Singapore. <https://ceur-ws.org/Vol-3534/paper5.pdf>
- W11 D. Garber, T. Burgstaller, A. Felfernig, V. M. Le, S. Lubos, **T. N. T. Tran**, and S. Polat-Erdeniz (2023). Collaborative Recommendation of Search Heuristics. In 25th International Workshop on Configuration (ConfWS 2023), Malaga, Spain. <https://ceur-ws.org/Vol-3509/paper6.pdf>
- W10 B. Gjergjizi, **T. N. T. Tran**, and A. Felfernig (2022). The Impacts of Primacy/Recency Effects on Item Review Sentiment Analysis. In *Proceedings of the 9th Joint Workshop on Interfaces and Human Decision Making for Recommender Systems (IntRS 2022)*, Hybrid Event, Seattle, USA. <https://ceur-ws.org/Vol-3222/paper3.pdf>
- W9 V. M. Le, **T. N. T. Tran**, and A. Felfernig (2022). Consistency-based Integration of Multi-Stakeholder Recommender Systems With Feature Model Configuration. In 26th ACM International Systems and Software Product Line Conference – Volume B (SPLC'22), Graz, Austria. <https://doi.org/10.1145/3503229.3547050>
- W8 V. M. Le, A. Felfernig, and **T. N. T. Tran** (2022). Test Case Aggregation for Efficient Feature Model Testing. In 26th ACM International Systems and Software Product Line Conference – Volume B (SPLC'22), Graz, Austria. <https://doi.org/10.1145/3503229.3547046>
- W7 **T. N. T. Tran**, C.I. Baumann, A. Felfernig, and V. M. Le (2021). The Immunity of Users' Item Selection from Serial Position Effects in Multi-Attribute Item Recommendation Scenarios. In *Proceedings of RecSys'21 Joint Workshop on Interfaces and Human Decision Making for Recommender Systems (IntRS'21)*, Amsterdam, Netherlands, pp. 101-111. <https://ceur-ws.org/Vol-2948/short2.pdf>
- W6 V. M. Le, **T. N. T. Tran**, M. Stettinger, L. Weißl, A. Felfernig, M. Atas, S.P. Erdeniz, and A. Popescu (2021). Counteracting Exam Cheating by Leveraging Configuration and Recommendation Techniques. In *Proceedings of the 23rd International Configuration Workshop*, Vienna, Austria, pp. 73-80. https://ceur-ws.org/Vol-2945/53-VML-ConfWS21_paper_7.pdf
- W5 A. Felfernig, V. M. Le, and **T. N. T. Tran** (2020). Supporting Feature Model-Based Configuration in Microsoft Excel. In *Proceedings of the 22nd International Configuration Workshop*, Vicenza, Italy, pp. 35-38.

- W4 M. Atas, **T. N. T. Tran**, R. Samer, A. Felfernig, and M. Stettinger (2018). Liquid Democracy in Group-based Configuration. In Proceedings of the 20th International Configuration Workshop, Graz, Austria, pp. 93-98. https://ceur-ws.org/Vol-2220/15_CONFWS18_paper_32.pdf (**ConfWS' 2018 Best Paper Award**)
- W3 **T. N. T. Tran**, M. Atas, M. Stettinger, and A. Felfernig (2016). An extension of CHOICLA User Interfaces for Configurable Products. In the 2nd Workshop on Recommender Systems and Big Data Analytics (RS-BDA'16), Graz, Austria, pp. 1-4. https://tugraz.elsevierpure.com/ws/portalfiles/portal/27357188/RS_BDA16_An_extension_of_CHOICLA_UI_for_configurable_products.pdf
- W2 A. Felfernig, M. Atas, **T. N. T. Tran**, and M. Stettinger (2016). Towards Group-based Configuration. In Proceedings of the 18th International Configuration Workshop, Toulouse, France, pp. 69-72. <https://asc.sai.tugraz.at/wp-content/uploads/sites/34/2014/01/group-based-configuration.pdf> (**ConfWS' 2016 Runner-up Best Paper Award**)
- W1 A. Felfernig, A. Shehadeh, M. Jeran, C. Gütl, **T. N. T. Tran**, M. Atas, S. Polat Eredeniz, M. Stettinger, A. Akcay, and S. Reiterer (2016). StudyBattles: A Learning Environment for Knowledge-based Configuration. In Proceedings of the 18th International Configuration Workshop, Toulouse, France, pp. 109-116. https://ksiresearch.org.ipage.com/seke/sekel7paper/sekel7paper_93.pdf (**SEKE' 2017 Best Paper Award**)

Other publications (Book chapters)

- B6 A. Felfernig, **T. N. T. Tran**, and V. M. Le (2021). Recommender Systems Beyond E-Commerce: Presence and Future. In: T. Dutta, M.K. Mandal (Eds.), Consumer Happiness: Multiple Perspectives. Studies in Rhythm Engineering. Springer, Singapore.
- B5 A. Felfernig, M. Atas, D. Helic, **T. N. T. Tran**, M. Stettinger, and R. Samer (2018). Algorithms for Group Recommendation. In A. Felfernig, L. Boratto, M. Stettinger, and M. Tkalcic (Eds.), Group Recommender Systems: An Introduction (pp. 27-58). SpringerBriefs in Electrical and Computer Engineering. Springer.
- B4 A. Felfernig, M. Atas, M. Stettinger, **T. N. T. Tran**, and S. Reiterer (2018). Group Recommender Applications. In A. Felfernig, L. Boratto, M. Stettinger, and M. Tkalcic (Eds.), Group Recommender Systems: An Introduction (pp. 75-89). SpringerBriefs in Electrical and Computer Engineering. Springer.
- B3 A. Felfernig, N. Tintarev, **T. N. T. Tran**, and M. Stettinger (2018). Explanations for Groups. In A. Felfernig, L. Boratto, M. Stettinger, and M. Tkalcic (Eds.), Group Recommender Systems: An Introduction (pp. 105-126). SpringerBriefs in Electrical and Computer Engineering. Springer.
- B2 A. Felfernig, M. Atas, R. Samer, M. Stettinger, **T. N. T. Tran**, and S. Reiterer (2018). Further Choice Scenarios. In A. Felfernig, L. Boratto, M. Stettinger, and M. Tkalcic (Eds.), Group Recommender Systems: An Introduction (pp. 129-144). SpringerBriefs in Electrical and Computer Engineering. Springer.
- B1 A. Felfernig, M. Atas, M. Stettinger, **T. N. T. Tran**, and G. Leitner (2018). Biases in Group Decisions. In A. Felfernig, L. Boratto, M. Stettinger, and M. Tkalcic (Eds.), Group Recommender Systems: An Introduction (pp. 145-155). SpringerBriefs in Electrical and Computer Engineering. Springer.

SKILLS

Programming languages: Java, Javascript, PHP, Python

Database: MySQL, SQL Server

Languages: Vietnamese (mother tongue), English (fluent), German (A2)

SCIENTIFIC COMMUNITY ACTIVITIES

Conference and Workshop Organization

- Local chair at SPLC' 2022 and CONFWS' 2022 (<https://2022.splc.net/committees/organization/>)

Program Committee Memberships and Reviewing

- UMAP' 2026
- CHI' 2023
- FinRec' 2022
- IntRS' 2022
- UMAP' 2022
- UMAP' 2021
- CHI' 2021
- Journal of Intelligent Information Systems (JIIS)

TEACHING

I am responsible for organizing and/or lecturing the following courses:

- **Recommender Systems** introduces the fundamental concepts and underlying algorithms of recommender systems, including content-based filtering, collaborative filtering with a focus on matrix factorization, hybrid recommendation, knowledge-based recommendation, and group recommendation approaches. In addition, the course covers related topics such as video recommendation, LLMs in recommender systems, explaining recommendations, evaluating recommender systems, and biases in recommender systems. Through this course, students gain both a solid theoretical foundation and an understanding of practical challenges in the design and assessment of recommender systems.
- **Configuration Systems** presents modern algorithms and methods that reflect the state of the art in configuration knowledge representation, interacting with configurators, as well as intelligent approaches to testing and debugging configuration knowledge bases.
- **Object-Oriented Analysis and Design** provides a foundational introduction to the core concepts and techniques of object-oriented software development. The course covers key topics such as requirements engineering, class diagrams, state charts, object-relational mapping, design patterns, automated testing, and fundamental software design principles. Through continuous hands-on exercises and practical assignments, students deepen their understanding, actively apply the learned concepts, and develop the skills needed to design well-structured and maintainable software systems.
- **Introduction to Structured Programming** provides a foundational overview of core programming concepts and techniques using a selected programming language as the primary vehicle for instruction.
- **Software Development Process** introduces a range of software process models and offers guidance on selecting and applying appropriate processes in different development contexts. The

course also covers essential aspects of high-quality software requirements, along with fundamental techniques for requirements prioritization and effort estimation.

- **Explanations in Artificial Intelligence** offers an introduction to the theoretical foundations of explainability in AI, with a particular focus on explanation approaches in both logic-based and sub-symbolic AI systems.

BACHELOR AND MASTER THESIS SUPERVISION

In addition to teaching, I also serve as a supervisor and co-supervisor for Bachelor's and Master's theses.

- Co-supervisor of the Master thesis on "*Application of Large Language Models in Usability Testing of Recommender Systems*", 2025. (Contact information: Naida Nozics, naida.nozic@student.tugraz.at)
- Co-supervisor of the Master thesis on "*Decision Biases in Food Recommendation Systems*", 2023 (Contact information: Amela Kurtic, amelakurtic.97@gmail.com)
- Supervisor of the Bachelor thesis on "*The Impacts of Privacy/Recency Effects on Item Review Sentiment Analysis*", published at IntRS'22: Joint Workshop on Interfaces and Human Decision Making for Recommender Systems, September 22, 2022, Seattle, US (hybrid event). (Contact information: Besnik Gjergjizi, besnik.gjergjizi@student.tugraz.at)
- Supervisor of the Bachelor thesis on "*The immunity of Users' Item Selection from Serial Position Effects in Multi-Attribute Item Recommendation Scenarios*", published at IntRS '21: Joint Workshop on Interfaces and Human Decision Making for Recommender Systems (2021), Amsterdam on September 25 or October 1, 2021. (Contact information: Carmen Baumann, carmen.baumann@student.tugraz.at)

PROJECT EXPERIENCES

- Participating in the WEWANT project, funded by the Austrian Research Promotion Agency (no. 850702), focusing on configuration technologies and decision-making support for user groups.
- Member of the PARXCEL project, dedicated to developing machine learning techniques to enhance constraint solving performance and prediction quality, alongside parallelized approaches for efficient analysis operations.
- Member of the GenRE project, contributing to the analysis of how Large Language Models can significantly improve requirements engineering tasks such as requirements definition, quality assurance, reuse, and prioritization.
- Member of the OpenSpace project, contributing to the development of novel methods for the automated analysis of software variability models and the generation of corresponding test cases, as well as machine learning-based techniques for identifying faulty software components and suboptimal parameter configurations that may lead to inefficiencies or erroneous system behavior.
- Serving as PI and leader for an FWF international project (under the WEAVE call) entitled "*HUMAX - Human Factors-Enhanced Explanations in Recommender Systems*", collaborating with Prof. Marko Tkalčič, a well-known researcher in the areas of Recommender Systems and Explanations from the Faculty of Mathematics, Natural Sciences and Information Technology, University of Primorska, Slovenia. The proposal was submitted in 2024 and, although not granted in the first round, it received very positive reviews (C2), highlighting its scientific merit and potential impact. We are currently revising the proposal based on the reviewers' feedback and preparing a resubmission.